

Zaid Khan

diaznaKh@gmail.com

[Zaid Khan](#)

[diaznaKh](#)

Education

National Institute of Technology Srinagar

2023 – Present

Bachelor of Technology in Chemical Engineering

Projects

6502 Microprocessor Emulator | C++

April 2025

- Built a cycle-accurate 6502 microprocessor emulator in C++, implementing CPU registers, memory interaction, instruction execution, and processor state management.
- Implemented the 6502 instruction set, addressing modes, status flags, stack operations, branching, and arithmetic/logic instructions.
- Designed a modular architecture separating CPU, memory, instruction decoding, and execution, improving maintainability and testability.

Redis Server | C++

January 2025

- Developed a minimal Redis server in C++ to understand the fundamentals of in-memory databases and client-server architecture.
- Implemented RESP protocol parsing, command execution, and basic persistence using multithreading and STL containers.

Maze Solver | C++, A*, Dijkstra, Graph Algorithms

March 2024

- Developed a grid-based maze solver in C++ by modeling maze cells as a graph and implementing shortest-path algorithms.
- Implemented Dijkstra's algorithm and A* with Manhattan-distance heuristics to find optimal paths between source and destination nodes.
- Compared search efficiency, nodes explored, path length, and execution time across different maze configurations.

Technical Skills

Languages: C, C++, Python

Databases: Redis, MySQL

Concepts: Object-Oriented Programming, Data Structures & Algorithms, Multithreading, Networking, Client-Server Architecture, System Design, Operating Systems

Technologies/Tools: Linux, Git, STL

Leadership / Extracurricular

Cursor Hackathon

March 2026

Lead Organiser

NIT Srinagar

- Led the planning and execution of a two-day national-level hackathon in Kashmir, attracting 600+ attendees including students, developers, and tech enthusiasts.
- Coordinated logistics, event operations, participant management, outreach, and on-ground execution across the two-day event.

Google Developer Student Club

Spring 2023 – Autumn 2024

Operations Lead

NIT Srinagar

- Organized workshops on Git, open-source, and software development best practices.

Campus Relations Coordinator

2026

Coordinator

NIT Srinagar

- Managed external campus relations and event coordination for student-focused initiatives.

Techvaganza

2024

Team Lead

NIT Srinagar

- Managed a 41-member team for 3 months, handling project execution and coordination.

Rang-e-Chinar

2025

Team Lead

NIT Srinagar

- Directed 2 project teams, ensuring timely execution and effective communication.